

LOGICAL REASONING (ANALYTICAL ABILITY)

Full Question Bank
Questions, Solutions, and Explanations

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SECTION: 01 Basic

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--- README.md ---

100 Logical Reasoning Practice Questions (Basic Level) with Solutions

This file contains 100 basic-level practice questions for Logical Reasoning, designed to help you prepare for fresher job aptitude tests, especially for AI/ML roles. Each question includes a detailed solution to build your foundational skills. Topics include series, analogies, coding-decoding, blood relations, directions, syllogisms, puzzles, data sufficiency, and non-verbal reasoning.

Series

1. Question: Find the next number in the series: 2, 4, 6, 8, ?

Solution:

The series is arithmetic with a common difference of 2: $2 + 2 = 4$, $4 + 2 = 6$, $6 + 2 = 8$.

Next number: $8 + 2 = 10$.

Answer: 10

2. Question: Find the next number: 1, 3, 9, 27, ?

Solution:

The series is geometric: each term is multiplied by 3 ($1 \times 3 = 3$, $3 \times 3 = 9$, $9 \times 3 = 27$).

Next number: $27 \times 3 = 81$.

Answer: 81

3. Question: Find the next number: 5, 10, 20, 40, ?

Solution:

Each term is multiplied by 2: $5 \times 2 = 10$, $10 \times 2 = 20$, $20 \times 2 = 40$.

Next number: $40 \times 2 = 80$.

Answer: 80

4. Question: Find the next number: 1, 4, 7, 10, ?

Solution:

Arithmetic series with a common difference of 3: $1 + 3 = 4$, $4 + 3 = 7$, $7 + 3 = 10$.

Next number: $10 + 3 = 13$.

Answer: 13

5. Question: Find the missing number: 2, 5, 11, ?, 26

Solution:

The series follows a pattern: $2 \times 2 + 1 = 5$, $5 \times 2 + 1 = 11$, $11 \times 2 + 1 = 23$, $23 \times 2 + 1 = 47$.

Missing number: 23.

Answer: 23

6. Question: Find the next number: 3, 6, 12, 24, ?

Solution:

Each term is multiplied by 2: $3 \times 2 = 6$, $6 \times 2 = 12$, $12 \times 2 = 24$.

Next number: $24 \times 2 = 48$.

Answer: 48

7. Question: Find the next number: 10, 8, 6, 4, ?

Solution:

Arithmetic series with a common difference of -2: $10 - 2 = 8$, $8 - 2 = 6$, $6 - 2 = 4$.

Next number: $4 - 2 = 2$.

Answer: 2

8. Question: Find the missing number: 1, 2, 4, 7, ?, 22

Solution:

Differences: $2 - 1 = 1$, $4 - 2 = 2$, $7 - 4 = 3$, next difference is 4, so $7 + 4 = 11$.

Next difference is 5: $11 + 5 = 16$, then 6: $16 + 6 = 22$.

Missing number: 11.

Answer: 11

9. Question: Find the next number: 1, 1, 2, 3, 5, ?

Solution:

Fibonacci series: each term is the sum of the two preceding ones ($1 + 1 = 2$, $1 + 2 = 3$, $2 + 3 = 5$).

Next number: $3 + 5 = 8$.

Answer: 8

10. Question: Find the next number: 4, 16, 64, 256, ?

Solution:

Each term is multiplied by 4: $4 \times 4 = 16$, $16 \times 4 = 64$, $64 \times 4 = 256$.

Next number: $256 \times 4 = 1024$.

Answer: 1024

Analogies

11. Question: Cat : Meow :: Dog : ?

Solution:

A cat's sound is 'meow'; a dog's sound is 'bark'.

Answer: Bark

12. Question: Pen : Write :: Scissors : ?

Solution:

A pen is used to write; scissors are used to cut.

Answer: Cut

13. Question: Big : Small :: Tall : ?

Solution:

Big is the opposite of small; tall is the opposite of short.

Answer: Short

14. Question: Apple : Fruit :: Carrot : ?

Solution:

Apple is a type of fruit; carrot is a type of vegetable.

Answer: Vegetable

15. Question: Book : Page :: Kitchen : ?

Solution:

A book contains pages; a kitchen contains utensils (or appliances, but utensils is more specific).

Answer: Utensils

16. Question: Sun : Day :: Moon : ?

Solution:

Sun is associated with day; moon is associated with night.

Answer: Night

17. Question: Teacher : Classroom :: Chef : ?

Solution:

A teacher works in a classroom; a chef works in a kitchen.

Answer: Kitchen

18. Question: Bird : Fly :: Fish : ?

Solution:

A bird moves by flying; a fish moves by swimming.

Answer: Swim

19. Question: Foot : Shoe :: Hand : ?

Solution:

A foot is covered by a shoe; a hand is covered by a glove.

Answer: Glove

20. Question: Car : Road :: Boat : ?

Solution:

A car travels on a road; a boat travels on water.

Answer: Water

Coding-Decoding

21. Question: If ?CAT? is coded as ?DBU,? how is ?DOG? coded?

Solution:

Each letter in CAT is shifted forward by 1 (C→D, A→B, T→U).

For DOG: D+1=E, O+1=P, G+1=H.

Answer: EPH

22. Question: If ?BOOK? is coded as ?CPPL,? how is ?READ? coded?

Solution:

Each letter is shifted forward by 1: B→C, O→P, O→P, K→L.

For READ: R+1=S, E+1=F, A+1=B, D+1=E.

Answer: SFBE

23. Question: If ?MAN? is coded as ?NBO,? how is ?PEN? coded?

Solution:

Each letter is shifted forward by 1: M→N, A→B, N→O.

For PEN: P+1=Q, E+1=F, N+1=O.

Answer: QFO

24. Question: If ?FISH? is coded as ?EJRI,? how is ?BIRD? coded?

Solution:

Each letter is shifted backward by 1: F→E, I→J, S→R, H→I.

For BIRD: B-1=A, I-1=H, R-1=Q, D-1=C.

Answer: AHQC

25. Question: If ?APPLE? is coded as ?BQQMF,? how is ?GRAPE? coded?

Solution:

Each letter is shifted forward by 1: A→B, P→Q, P→Q, L→M, E→F.

For GRAPE: G+1=H, R+1=S, A+1=B, P+1=Q, E+1=F.

Answer: HSBQF

26. Question: If ?SUN? is coded as 54, how is ?MOON? coded?

Solution:

SUN: S=19, U=21, N=14; sum = 19+21+14 = 54.

MOON: M=13, O=15, O=15, N=14; sum = 13+15+15+14 = 57.

Answer: 57

27. Question: If ?DOG? is coded as ?WLT,? how is ?CAT? coded?

Solution:

Each letter is shifted backward by 1 in reverse order: D→C (W), O→N (L), G→F (T).

For CAT: Reverse = TAC; T-1=S, A-1=Z, C-1=B. Code = SZB.

Answer: SZB

28. Question: If ?STAR? is coded as ?TUBS,? how is ?MOON? coded?

Solution:

Each letter is shifted forward by 1: S→T, T→U, A→B, R→S.

For MOON: M+1=N, O+1=P, O+1=P, N+1=O.

Answer: NPPO

29. Question: If ?COW? is coded as 24, how is ?PIG? coded?

Solution:

COW: C=3, O=15, W=23; sum = $3+15+23 = 41$ (but given 24, assume positional sum or typo).

Assume sum of positions: P=16, I=9, G=7; sum = $16+9+7 = 32$.

Answer: 32

30. Question: If ?BALL? is coded as ?CBMM,? how is ?GAME? coded?

Solution:

Each letter is shifted forward by 1: B?C, A?B, L?M, L?M.

For GAME: G+1=H, A+1=B, M+1=N, E+1=F.

Answer: HBNF

Blood Relations

31. Question: A is B's brother, and C is B's sister. If D is C's mother, who is D to A?

Solution:

A and B are brothers, C is their sister, D is C's mother.

D is the mother of A, B, and C.

Answer: Mother

32. Question: If X is Y's father, and Y is Z's sister, who is X to Z?

Solution:

X is the father of Y and Z (siblings).

Answer: Father

33. Question: P is Q's son, and R is Q's wife. Who is R to P?

Solution:

R is Q's wife, and P is their son.

R is P's mother.

Answer: Mother

34. Question: A is B's father, and B is C's daughter. Who is C to A?

Solution:

A is the father of B, and B is C's daughter.

C is A's wife.

Answer: Wife

35. Question: If M is N's sister, and O is N's father, who is O to M?

Solution:

M and N are sisters, O is N's father.

O is M's father.

Answer: Father

36. Question: X is Y's brother, and Z is Y's mother. Who is Z to X?

Solution:

Z is the mother of Y and X (siblings).

Answer: Mother

37. Question: If A is B's son, and C is B's sister, who is C to A?

Solution:

A is B's son, and C is B's sister.

C is A's mother.

Answer: Mother

38. Question: P is Q's father, and R is Q's daughter. Who is R to P?

Solution:

R is Q's daughter, and P is Q's father.

R is P's daughter.

Answer: Daughter

39. Question: If X is Y's sister, and Z is Y's father, who is Z to X?

Solution:

X and Y are sisters, Z is Y's father.

Z is X's father.

Answer: Father

40. Question: A is B's mother, and B is C's sister. Who is A to C?

Solution:

A is B's mother, and B is C's sister.

A is C's mother.

Answer: Mother

Directions

41. Question: A man walks 5 km north, then 3 km east. How far is he from the starting point?

Solution:

Forms a right triangle. North = 5 km, East = 3 km.

Distance = $\sqrt{5^2 + 3^2} = \sqrt{25 + 9} = \sqrt{34} \approx 5.83$ km.

Answer: 5.83 km

42. Question: A person walks 4 km south, then 3 km west. In which direction is he from the starting point?

Solution:

South and west form a right angle. He is southwest of the starting point.

Answer: Southwest

43. Question: A man walks 6 km east, then 8 km north. How far is he from the starting point?

Solution:

Distance = $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km.

Answer: 10 km

44. Question: A person walks 2 km north, then 2 km west, then 2 km south. Where is he relative to the starting point?

Solution:

North 2 km, south 2 km cancel out. He moves 2 km west.

Answer: 2 km west

45. Question: A man walks 5 km south, then 5 km east, then 5 km north. Where is he from the starting point?

Solution:

South 5 km, north 5 km cancel out. He moves 5 km east.

Answer: 5 km east

46. Question: A person walks 10 km west, then 10 km north. What is the shortest distance to the starting point?

Solution:

Distance = $\sqrt{10^2 + 10^2} = \sqrt{200} \approx 14.14$ km.

Answer: 14.14 km

47. Question: A man walks 3 km north, then 4 km east, then 3 km south. Where is he from the starting point?

Solution:

North 3 km, south 3 km cancel out. He moves 4 km east.

Answer: 4 km east

48. Question: A person walks 6 km east, then 8 km south, then 6 km west. Where is he from the starting point?

Solution:

East 6 km, west 6 km cancel out. He moves 8 km south.

Answer: 8 km south

49. Question: A man walks 7 km north, then 24 km east. How far is he from the starting point?

Solution:

Distance = $\sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$ km.

Answer: 25 km

50. Question: A person walks 4 km west, then 3 km north. In which direction is he from the starting point?

Solution:

West and north form a right angle. He is northwest of the starting point.

Answer: Northwest

Syllogisms

51. Question: All cats are mammals. All mammals are animals. Conclusion: All cats are animals.

Solution:

All cats are mammals (A?B), all mammals are animals (B?C).

Therefore, all cats are animals (A?C). Valid.

Answer: True

52. Question: Some dogs are pets. All pets are friendly. Conclusion: Some dogs are friendly.

Solution:

Some dogs are pets (Some A are B), all pets are friendly (B?C).

Some dogs are friendly (Some A are C). Valid.

Answer: True

53. Question: All birds fly. Some sparrows are birds. Conclusion: Some sparrows fly.

Solution:

All birds fly (A?B), some sparrows are birds (Some C are A).

Some sparrows fly (Some C are B). Valid.

Answer: True

54. Question: No fish are mammals. All whales are mammals. Conclusion: No whales are fish.

Solution:

No fish are mammals (No A are B), all whales are mammals (C?B).

No whales are fish (No C are A). Valid.

Answer: True

55. Question: Some apples are red. Some red things are fruits. Conclusion: Some apples are fruits.

Solution:

Some apples are red (Some A are B), some red things are fruits (Some B are C).

No definite conclusion about apples and fruits. Invalid.

Answer: False

56. Question: All roses are flowers. Some flowers are red. Conclusion: Some roses are red.

Solution:

All roses are flowers (A?B), some flowers are red (Some B are C).

No definite conclusion. Invalid.

Answer: False

57. Question: No cats are dogs. All dogs are pets. Conclusion: No cats are pets.

Solution:

No cats are dogs (No A are B), all dogs are pets (B?C).

No definite conclusion. Invalid.

Answer: False

58. Question: All engineers are smart. Some smart people are coders. Conclusion: Some engineers are coders.

Solution:

All engineers are smart (A?B), some smart people are coders (Some B are C).

No definite conclusion. Invalid.

Answer: False

59. Question: All books are pages. All pages are paper. Conclusion: All books are paper.

Solution:

All books are pages (A?B), all pages are paper (B?C).

All books are paper (A?C). Valid.

Answer: True

60. Question: Some cars are fast. All fast things are expensive. Conclusion: Some cars are expensive.

Solution:

Some cars are fast (Some A are B), all fast things are expensive (B?C).

Some cars are expensive (Some A are C). Valid.

Answer: True

Puzzles

61. Question: Three houses A, B, C are in a row. A is to the left of B, and C is to the right of A. Which house is in the middle?

Solution:

A is left of B, C is right of A. Order: A, B, C.

B is in the middle.

Answer: B

62. Question: In a race, P finishes before Q but after R. Who finishes first?

Solution:

R finishes before P, P before Q. Order: R, P, Q.

R finishes first.

Answer: R

63. Question: Four people A, B, C, D sit in a row. A is next to B, and C is next to D. Who can be at the ends?

Solution:

A and B are together, C and D are together. Possible orders: A-B-C-D, D-C-B-A, etc.

Ends can be A or D (or B or C in reverse).

Answer: A or D

64. Question: Three friends X, Y, Z have different ages. X is younger than Y, and Z is older than X. Who is the youngest?

Solution:

X is younger than Y, Z is older than X.

X is the youngest.

Answer: X

65. Question: In a line, P is second from the left, and Q is third from the right. If there are 5 people, who is in the middle?

Solution:

Total 5 positions. P is 2nd (left). Q is 3rd from right, so 3rd (5-3+1).

Order: _ P _ Q _. Middle is 3rd, which is Q.

Answer: Q

66. Question: A, B, C are three boxes. A is heavier than B, and B is heavier than C. Which is the lightest?

Solution:

$A > B > C$.

C is the lightest.

Answer: C

67. Question: In a class, M is taller than N but shorter than O. Who is the tallest?

Solution:

$O > M > N$.

O is the tallest.

Answer: O

68. Question: Three cars P, Q, R are parked. P is to the left of Q, and R is to the right of Q. Which is in the middle?

Solution:

Order: P, Q, R.

Q is in the middle.

Answer: Q

69. Question: Four students A, B, C, D sit in a circle. A is opposite B, and C is next to A. Who is opposite C?

Solution:

A and B are opposite. C is next to A. Possible order: A-C-D-B.

Opposite C is D.

Answer: D

70. Question: In a queue, X is third from the front, and Y is second from the back. If there are 6 people, who is between X and Y?

Solution:

X is 3rd. Y is 5th (6-2+1). Order: _ _ X _ Y _.

Person in 4th position is between X and Y.

Answer: 4th person

Data Sufficiency

71. Question: Is $x > 5$?

(1) $x + 2 = 7$

(2) $x - 3 = 2$

Solution:

(1) $(x + 2 = 7)$, so $(x = 5)$. Not sufficient ($x = 5$, not > 5).

(2) $(x - 3 = 2)$, so $(x = 5)$. Not sufficient.

Both give $(x = 5)$. Not sufficient.

Answer: Not sufficient

72. Question: What is the value of y ?

(1) $y + 4 = 10$

(2) $2y = 12$

Solution:

(1) $(y + 4 = 10)$, so $(y = 6)$. Sufficient.

(2) $(2y = 12)$, so $(y = 6)$. Sufficient.

Answer: Each alone is sufficient

73. Question: Is a rectangle a square?

(1) All sides are equal

(2) One side is 5

Solution:

(1) All sides equal implies a square. Sufficient.

(2) One side = 5 does not confirm equal sides. Not sufficient.

Answer: Statement 1 alone is sufficient

74. Question: What is the age of A?

(1) A is twice as old as B

(2) B is 10 years old

Solution:

(1) $A = 2B$. Not sufficient without B's age.

(2) $B = 10$. Not sufficient without A's relation.

Both: $A = 2 \times 10 = 20$. Sufficient.

Answer: Both statements together are sufficient

75. Question: Is x positive?

(1) $x^2 = 16$

(2) $x > 0$

Solution:

(1) $(x^2 = 16)$, so $(x = \pm 4)$. Not sufficient.

(2) $(x > 0)$. Sufficient.

Answer: Statement 2 alone is sufficient

76. Question: What is the value of z ?

(1) $z + 5 = 8$

(2) $z - 2 = 1$

Solution:

(1) $(z + 5 = 8)$, so $(z = 3)$. Sufficient.

(2) $(z - 2 = 1)$, so $(z = 3)$. Sufficient.

Answer: Each alone is sufficient

77. Question: Is a triangle equilateral?

(1) Two sides are equal

(2) One angle is 60°

Solution:

(1) Two sides equal implies isosceles, not necessarily equilateral. Not sufficient.

(2) One angle = 60° does not confirm all angles equal. Not sufficient.

Both: Still not sufficient without all sides or angles equal.

Answer: Not sufficient

78. Question: What is the value of $x + y$?

(1) $x = 5$

(2) $y = 3$

Solution:

(1) $x = 5$. Not sufficient without y .

(2) $y = 3$. Not sufficient without x .

Both: $x + y = 5 + 3 = 8$. Sufficient.

Answer: Both statements together are sufficient

79. Question: Is x even?

(1) x is divisible by 2

(2) x is a multiple of 3

Solution:

(1) Divisible by 2 means x is even. Sufficient.

(2) Multiple of 3 (e.g., 3, 6) may or may not be even. Not sufficient.

Answer: Statement 1 alone is sufficient

80. Question: What is the value of p ?

(1) $p + q = 10$

(2) $p - q = 4$

Solution:

Add (1) and (2): $(p + q) + (p - q) = 10 + 4$, so $(2p = 14)$, $(p = 7)$.

Answer: Both statements together are sufficient

Non-Verbal Reasoning

81. Question: Which shape completes the pattern: Square, Circle, Square, Circle, ?

Solution:

The pattern alternates: Square, Circle, Square, Circle.

Next is Square.

Answer: Square

82. Question: If a figure rotates 90° clockwise, and it's a triangle pointing up, what is the new orientation?

Solution:

Upward triangle rotates 90° clockwise to point right.

Answer: Right

83. Question: In a 3×3 grid, top row: 1, 2, 3; middle row: 4, 5, 6; bottom row: 7, 8, ?. What is the missing number?

Solution:

Sequential numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9.

Missing number: 9.

Answer: 9

84. Question: A pattern shows 2 stars, 4 stars, 6 stars. How many stars in the next term?

Solution:

Arithmetic sequence: 2, 4, 6 (common difference 2).

Next: $6 + 2 = 8$.

Answer: 8 stars

85. Question: If a square is divided into 4 smaller squares, and 3 are shaded, how many are unshaded?

Solution:

Total squares = 4. Shaded = 3. Unshaded = $4 - 3 = 1$.

Answer: 1

86. Question: A clock shows 3:00. After rotating 180° , what time does it show?

Solution:

180° rotation flips the clock. 3:00 becomes 9:00 (opposite on clock face).

Answer: 9:00

87. Question: Which shape follows: Circle, Triangle, Square, Circle, ?

Solution:

Pattern repeats every 3: Circle, Triangle, Square.

After Circle, next is Triangle.

Answer: Triangle

88. Question: In a 2×2 grid: top-left = 1, top-right = 2, bottom-left = 3, bottom-right = ?. What is the missing number?

Solution:

Sequential: 1, 2, 3, 4.

Missing number: 4.

Answer: 4

89. Question: A pattern shows 1 dot, 3 dots, 5 dots. How many dots in the next term?

Solution:

Odd numbers: 1, 3, 5. Next: $5 + 2 = 7$.

Answer: 7 dots

90. Question: If a cube is painted on all faces and cut into 8 smaller cubes, how many have exactly 2 faces painted?

Solution:

A $2 \times 2 \times 2$ cube has 8 smaller cubes. Cubes with 2 faces painted are edge cubes (not corners).

Each edge has 1 such cube, and there are 12 edges.

Answer: 12

Additional Basic Questions

91. Question: Find the next letter: A, C, E, G, ?

Solution:

Letters correspond to positions: A=1, C=3, E=5, G=7 (odd numbers).

Next: 9th letter = I.

Answer: I

92. Question: If ?HOUSE? is coded as ?IPVTF,? how is ?GARDEN? coded?

Solution:

Each letter in HOUSE is shifted forward by 1: H?I, O?P, U?V, S?T, E?F.

For GARDEN: G+1=H, A+1=B, R+1=S, D+1=E, E+1=F, N+1=O.

Answer: HSBJFO

93. Question: A is B?s father, and C is B?s sister. Who is A to C?

Solution:

A is B?s father, C is B?s sister.

A is C?s father.

Answer: Father

94. Question: A person walks 3 km north, then 4 km west. How far is he from the starting point?

Solution:

Distance = $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$ km.

Answer: 5 km

95. Question: All pens are tools. Some tools are sharp. Conclusion: Some pens are sharp.

Solution:

All pens are tools (A?B), some tools are sharp (Some B are C).

No definite conclusion???

System: conclusion. Invalid.

Answer: False

96. Question: Find the next number: 7, 14, 28, 56, ?

Solution:

Each term is multiplied by 2: $7 \times 2 = 14$, $14 \times 2 = 28$, $28 \times 2 = 56$.

Next number: $56 \times 2 = 112$.

Answer: 112

97. Question: If ?RAT? is coded as ?SBU,? how is ?COW? coded?

Solution:

Each letter is shifted forward by 1: R?S, A?B, T?U.

For COW: C+1=D, O+1=P, W+1=X.

Answer: DPX

98. Question: A man walks 5 km east, then 12 km north. How far is he from the starting point?

Solution:

Distance = $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ km.

Answer: 13 km

99. Question: Some cats are black. All black things are dark. Conclusion: Some cats are dark.

Solution:

Some cats are black (Some A are B), all black things are dark (B?C).

Some cats are dark (Some A are C). Valid.

Answer: True

100. Question: In a 3x3 grid, top row: 2, 4, 6; middle row: 8, 10, 12; bottom row: 14, 16, ?. What is the missing number?

Solution:

Sequential even numbers: 2, 4, 6, 8, 10, 12, 14, 16, 18.

Missing number: 18.

Answer: 18

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SECTION: 02 Intermediate
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100 Logical Reasoning Practice Questions (Intermediate Level) with Solutions

This file contains 100 intermediate-level practice questions for Logical Reasoning, designed to enhance your preparation for fresher job aptitude tests, especially for AI/ML roles. Each question includes a detailed solution to strengthen your problem-solving skills. Topics include series, analogies, coding-decoding, blood relations, directions, syllogisms, puzzles, data sufficiency, critical reasoning, and non-verbal reasoning.

Series

1. Question: Find the next number in the series: 2, 5, 11, 23, ?

Solution:

The pattern is: each term is obtained by doubling the previous term and adding 1.

$$2 \times 2 + 1 = 5, 5 \times 2 + 1 = 11, 11 \times 2 + 1 = 23, 23 \times 2 + 1 = 47.$$

Answer: 47

2. Question: Find the missing number: 3, 7, 16, ?, 48

Solution:

The pattern is: multiply by 2 and add 1, then multiply by 2 and add 2, and so on.

$$3 \times 2 + 1 = 7, 7 \times 2 + 2 = 16, 16 \times 2 + 3 = 35, 35 \times 2 + 4 = 74.$$

Check: 35 fits as $16 \times 2 + 3 = 35$, and $35 \times 2 + 6 = 76$ (adjust pattern).

$$\text{Correct pattern: } 3 \times 2 + 1 = 7, 7 \times 2 + 2 = 16, 16 \times 2 + 3 = 35, 35 \times 2 + 4 = 74.$$

Missing number: 35.

Answer: 35

3. Question: Find the next number: 1, 4, 10, 22, ?

Solution:

$$\text{Differences: } 4 - 1 = 3, 10 - 4 = 6, 22 - 10 = 12.$$

$$\text{Differences double: } 3 \times 2 = 6, 6 \times 2 = 12, 12 \times 2 = 24.$$

$$\text{Next number: } 22 + 24 = 46.$$

Answer: 46

4. Question: Find the next number: 8, 6, 9, 23, ?

Solution:

Pattern: multiply by 1 and subtract 2, multiply by 2 and subtract 3, multiply by 3 and subtract 4, multiply by 4 and subtract 5.

$$8 \times 1 - 2 = 6, 6 \times 1.5 - 3 = 9, 9 \times 2.5 - 4 = 22.5 \text{ ? } 23, 23 \times 3.5 - 5 = 75.5 \text{ ? } 76.$$

Answer: 76

5. Question: Find the missing number: 2, 6, 12, ?, 30

Solution:

Pattern: multiply by 3, multiply by 2, multiply by 1.5, multiply by 1.25.

$$2 \times 3 = 6, 6 \times 2 = 12, 12 \times 1.5 = 18, 18 \times 1.25 = 22.5 \text{ (adjust).}$$

Correct: 2, 6, 12, 18, 30. Missing number: 18.

Answer: 18

6. Question: Find the next number: 5, 10, 25, 80, ?

Solution:

Pattern: multiply by 2, multiply by 2.5, multiply by 3.2, multiply by 4.

$$5 \times 2 = 10, 10 \times 2.5 = 25, 25 \times 3.2 = 80, 80 \times 4 = 320.$$

Answer: 320

7. Question: Find the next number: 1, 3, 8, 22, ?

Solution:

Pattern: multiply by 2 and add 1, multiply by 2 and add 2, multiply by 2 and add 6, multiply by 2 and add 12.

$$1 \times 2 + 1 = 3, 3 \times 2 + 2 = 8, 8 \times 2 + 6 = 22, 22 \times 2 + 12 = 56.$$

Answer: 56

8. Question: Find the missing number: 4, 9, 25, ?, 121

Solution:

The series is: 2^2 , 3^2 , 5^2 , 7^2 , 11^2 (squares of prime numbers).

Missing number: $7^2 = 49$.

Answer: 49

9. Question: Find the next number: 7, 14, 28, 49, ?

Solution:

Pattern: differences are 7, 14, 21 (multiples of 7).

Next difference: 28.

$$49 + 28 = 77.$$

Answer: 77

10. Question: Find the next number: 2, 8, 20, 38, ?

Solution:

Differences: 6, 12, 18. Pattern: add 6, add 12, add 18, add 24.

Next number: $38 + 24 = 62$.

Answer: 62

Analogies

11. Question: Clock : Time :: Thermometer : ?

Solution:

A clock measures time; a thermometer measures temperature.

Answer: Temperature

12. Question: Square : Four :: Pentagon : ?

Solution:

A square has four sides; a pentagon has five sides.

Answer: Five

13. Question: Tree : Forest :: Soldier : ?

Solution:

A tree is part of a forest; a soldier is part of an army.

Answer: Army

14. Question: Key : Lock :: Password : ?

Solution:

A key opens a lock; a password opens a system (or account).

Answer: System

15. Question: Doctor : Hospital :: Lawyer : ?

Solution:

A doctor works in a hospital; a lawyer works in a courtroom (or law firm, but courtroom is more specific).

Answer: Courtroom

16. Question: Rain : Umbrella :: Sun : ?

Solution:

An umbrella protects from rain; sunglasses (or sunscreen) protect from the sun.

Answer: Sunglasses

17. Question: Engine : Car :: Heart : ?

Solution:

An engine powers a car; a heart powers a body.

Answer: Body

18. Question: Map : Geography :: Calculator : ?

Solution:

A map is a tool for geography; a calculator is a tool for mathematics.

Answer: Mathematics

19. Question: Seed : Plant :: Egg : ?

Solution:

A seed grows into a plant; an egg grows into a bird (or animal, but bird is common).

Answer: Bird

20. Question: Compass : Direction :: Scale : ?

Solution:

A compass measures direction; a scale measures weight.

Answer: Weight

Coding-Decoding

21. Question: If ?MANGO? is coded as ?OCPIQ,? how is ?APPLE? coded?

Solution:

Each letter in MANGO is shifted forward by 2: $M+2=O$, $A+2=C$, $N+2=P$, $G+2=I$, $O+2=Q$.

For APPLE: $A+2=C$, $P+2=R$, $P+2=R$, $L+2=N$, $E+2=G$.

Answer: CRRNG

22. Question: If ?STAR? is coded as ?UVYZ,? how is ?MOON? coded?

Solution:

Each letter is shifted forward by 2, and the word is reversed: $S+2=U$, $T+2=V$, $A+2=Y$, $R+2=Z$, reverse = ZYXW.

For MOON: $M+2=O$, $O+2=Q$, $O+2=Q$, $N+2=P$. Reverse QPPO: OPPQ.

Answer: OPPQ

23. Question: If ?BOOK? is coded as ?APPL,? how is ?READ? coded?

Solution:

Each letter is shifted backward by 1: B-1=A, O-1=N, O-1=N, K-1=J (adjust to APPL).

Correct: B-1=A, O-1=N, O-1=N, K-1=J, but given APPL, check pattern.

Assume typo, standard shift: R-1=Q, E-1=D, A-1=Z, D-1=C.

Answer: QDZC

24. Question: If ?FISH? is coded as ?IRTK,? how is ?BIRD? coded?

Solution:

Each letter is shifted forward by 2: F+2=I, I+2=K, S+2=U, H+2=J (adjust to IRTK).

For BIRD: B+2=D, I+2=K, R+2=T, D+2=F.

Answer: DKTF

25. Question: If ?COW? is coded as ?XLF,? how is ?PIG? coded?

Solution:

Reverse COW: WOC. Shift backward by 1: W-1=V, O-1=N, C-1=B. Reverse: BNV (adjust to XLF).

Correct: C+2=X, O+2=Q, W+2=Y, reverse: YQX ? XLF pattern.

For PIG: P+2=R, I+2=K, G+2=I. Reverse: IKR ? RKI.

Answer: RKI

26. Question: If ?SUN? is coded as ?72,? how is ?MOON? coded?

Solution:

SUN: S=19, U=21, N=14; product = $19 \times 21 \times 14 = 5586$, sum digits = $5+5+8+6 = 24$ (adjust to 72).

Assume positional: M=13, O=15, O=15, N=14; product = $13 \times 15 \times 15 \times 14 = 40950$, sum digits = 18.

Answer: 18

27. Question: If ?DOG? is coded as ?EPL,? how is ?CAT? coded?

Solution:

Each letter is shifted forward by 1: D+1=E, O+1=P, G+1=H (adjust to EPL).

For CAT: C+1=D, A+1=B, T+1=U.

Answer: DBU

28. Question: If ?APPLE? is coded as ?ZKKAF,? how is ?GRAPE? coded?

Solution:

Each letter is shifted backward by 1, and the word is reversed: A-1=Z, P-1=O, P-1=O, L-1=K, E-1=D, reverse = DKOOZ ? ZKKAF.

For GRAPE: G-1=F, R-1=Q, A-1=Z, P-1=O, E-1=D. Reverse: DOZQF.

Answer: DOZQF

29. Question: If ?BALL? is coded as ?AFOT,? how is ?GAME? coded?

Solution:

Pattern: B-2=Z, A-2=Y, L-2=J, L-2=J, adjust to AFOT.

Correct: B+1=C-2=A, A+1=B-2=Z, L+1=M-2=K, L+1=M-2=K ? AFOT-like.

For GAME: G+1=H-2=F, A+1=B-2=Z, M+1=N-2=L, E+1=F-2=D.

Answer: FZLD

30. Question: If ?RAT? is coded as ?UCS,? how is ?COW? coded?

Solution:

Each letter is shifted forward by 3: R+3=U, A+3=D, T+3=W (adjust to UCS).

For COW: C+3=F, O+3=R, W+3=Z.

Answer: FRZ

Blood Relations

31. Question: A is B's father, C is B's sister, and D is A's wife. If E is C's son, who is E to B?

Solution:

A (father) and D (wife) have children B and C (sisters). E is C's son.

Since B and C are siblings, E (C's son) is B's son.

Answer: Son

32. Question: P is Q's brother, R is Q's mother, and S is P's father. Who is R to S?

Solution:

S (father) and R (mother) have children P and Q (siblings).

R is S's wife.

Answer: Wife

33. Question: X is Y's sister, Z is Y's father, and W is X's mother. Who is W to Z?

Solution:

Z (father) and W (mother) have children X and Y (sisters).

W is Z's wife.

Answer: Wife

34. Question: A is B's son, C is B's mother, and D is A's sister. Who is C to D?

Solution:

C (mother) and B (father) have children A (son) and D (sister).

C is D's mother.

Answer: Mother

35. Question: M is N's father, O is N's sister, and P is M's wife. If Q is O's daughter, who is Q to M?

Solution:

M (father) and P (wife) have children N and O (siblings). Q is O's daughter.

Q is M's daughter.

Answer: Daughter

36. Question: If X is Y's brother, Z is Y's father, and W is X's mother, who is Y to W?

Solution:

W (mother) and Z (father) have children X and Y (brothers).

Y is W's son.

Answer: Son

37. Question: A is B's wife, C is B's son, and D is C's sister. Who is D to A?

Solution:

A (wife) and B (husband) have children C (son) and D (sister).

D is A's daughter.

Answer: Daughter

38. Question: P is Q's father, R is Q's sister, and S is P's wife. If T is R's son, who is T to S?

Solution:

P (father) and S (wife) have children Q and R (siblings). T is R's son.

T is S's son.

Answer: Son

39. Question: If X is Y's sister, Z is Y's mother, and W is X's father, who is Y to W?

Solution:

W (father) and Z (mother) have children X and Y (sisters).

Y is W's daughter.

Answer: Daughter

40. Question: A is B's father, C is B's daughter, and D is C's mother. Who is D to A?

Solution:

A (father) and D (mother) have daughter C. B is C's father, so B = A.

D is A's wife.

Answer: Wife

Directions

41. Question: A man walks 4 km north, 3 km east, 4 km south, and 5 km west. Where is he from the starting point?

Solution:

North 4 km and south 4 km cancel out. East 3 km and west 5 km: net 2 km west.

He is 2 km west of the starting point.

Answer: 2 km west

42. Question: A person walks 5 km south, 6 km west, 5 km north, and 2 km east. How far is he from the starting point?

Solution:

South 5 km and north 5 km cancel out. West 6 km and east 2 km: net 4 km west.

Distance = 4 km.

Answer: 4 km

43. Question: A man starts at point A, walks 8 km east to B, 6 km north to C, 8 km west to D. Where is D relative to A?

Solution:

East 8 km and west 8 km cancel out. Net: 6 km north.

D is 6 km north of A.

Answer: 6 km north

44. Question: A person walks 10 km north, 5 km east, turns 90° left, walks 5 km, then turns 90° right, walks 5 km. Where is he now?

Solution:

North 10 km, east 5 km. 90° left (faces north), walks 5 km north (total 15 km north).

90° right (faces east), walks 5 km east (total 10 km east).

Distance = $\sqrt{15^2 + 10^2} = \sqrt{225 + 100} = \sqrt{325} \approx 18.03$ km.

Answer: 18.03 km northeast

45. Question: A man walks 3 km west, 4 km south, 5 km east, 4 km north. Where is he from the starting point?

Solution:

West 3 km and east 5 km: net 2 km east. South 4 km and north 4 km cancel out.

He is 2 km east.

Answer: 2 km east

46. Question: A person walks 6 km east, turns 90° right, walks 8 km, turns 90° right, walks 6 km. Where is he now?

Solution:

East 6 km, 90° right (faces south), walks 8 km south, 90° right (faces west), walks 6 km west.

Net: 0 km east/west, 8 km south.

He is 8 km south.

Answer: 8 km south

47. Question: A man walks 7 km north, 24 km east, 7 km south, 12 km west. How far is he from the starting point?

Solution:

North 7 km and south 7 km cancel out. East 24 km and west 12 km: net 12 km east.

Distance = 12 km.

Answer: 12 km

48. Question: A person starts at O, walks 5 km west to P, 5 km north to Q, 5 km east to R, 5 km south to S. Where is S relative to O?

Solution:

West 5 km and east 5 km cancel out. North 5 km and south 5 km cancel out.

S is at O.

Answer: At starting point

49. Question: A man walks 4 km north, turns 135° clockwise, walks 4 km. How far is he from the starting point?

Solution:

North 4 km. 135° clockwise from north (0°) = 135° (southeast).

Walks 4 km southeast: $x = 4 \cos 45^\circ = 4/\sqrt{2} \approx 2.83$ km east, $y = 4 \sin 45^\circ = 4/\sqrt{2} \approx 2.83$ km south.

Total: 0 km east + 2.83 km east = 2.83 km east; 4 km north - 2.83 km south = 1.17 km north.

Distance = $\sqrt{2.83^2 + 1.17^2} \approx \sqrt{8 + 1.37} \approx 3.06$ km.

Answer: 3.06 km

50. Question: A person walks 3 km east, 4 km north, turns 90° left, walks 5 km. Where is he from the starting point?

Solution:

East 3 km, north 4 km. 90° left from north (faces west), walks 5 km west.

Net: 3 km east - 5 km west = 2 km west, 4 km north.

Distance = $\sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.47$ km.

Answer: 4.47 km northwest

Syllogisms

51. Question: Some A are B. All B are C. No C are D. Conclusion: No A are D.

Solution:

Some A are B (Some A are B), all B are C (B?C), no C are D (No C are D).

Some A are C, but no C are D, so no A (that are C) are D.

Conclusion may not hold for all A. Invalid.

Answer: False

52. Question: All X are Y. Some Y are Z. No Z are W. Conclusion: No X are W.

Solution:

All X are Y (X?Y), some Y are Z (Some Y are Z), no Z are W (No Z are W).

Some Y are Z, but no Z are W. No definite conclusion about X and W. Invalid.

Answer: False

53. Question: No P are Q. All Q are R. Some R are S. Conclusion: Some S are not P.

Solution:

No P are Q (No P are Q), all Q are R (Q?R), some R are S (Some R are S).

No P are R (since no P are Q and Q?R). Some R are S, so some S are not P. Valid.

Answer: True

54. Question: All M are N. Some N are O. All O are P. Conclusion: Some M are P.

Solution:

All M are N (M?N), some N are O (Some N are O), all O are P (O?P).

Some N are O, and all O are P, so some N are P. Since all M are N, some M are P. Valid.

Answer: True

55. Question: Some A are B. No B are C. All C are D. Conclusion: No A are D.

Solution:

Some A are B (Some A are B), no B are C (No B are C), all C are D (C?D).

No A (that are B) are C, but no conclusion about A and D. Invalid.

Answer: False

56. Question: All X are Y. No Y are Z. Some Z are W. Conclusion: No X are W.

Solution:

All X are Y (X?Y), no Y are Z (No Y are Z), some Z are W (Some Z are W).

No X are Z (since X?Y and no Y are Z). No conclusion about X and W. Invalid.

Answer: False

57. Question: Some P are Q. All Q are R. No R are S. Conclusion: Some P are not S.

Solution:

Some P are Q (Some P are Q), all Q are R (Q?R), no R are S (No R are S).

Some P are R, and no R are S, so some P are not S. Valid.

Answer: True

58. Question: No M are N. Some N are O. All O are P. Conclusion: Some P are not M.

Solution:

No M are N (No M are N), some N are O (Some N are O), all O are P (O?P).

Some O are P, and no M are N, so some P are not M. Valid.

Answer: True

59. Question: All A are B. No B are C. Some C are D. Conclusion: Some A are D.

Solution:

All A are B (A?B), no B are C (No B are C), some C are D (Some C are D).

No A are C, so no conclusion about A and D. Invalid.

Answer: False

60. Question: Some X are Y. All Y are Z. No Z are W. Conclusion: Some X are not W.

Solution:

Some X are Y (Some X are Y), all Y are Z (Y?Z), no Z are W (No Z are W).

Some X are Z, and no Z are W, so some X are not W. Valid.

Answer: True

Puzzles

61. Question: Four people A, B, C, D sit in a row. A is not next to B. C is to the left of D. Who can be at the ends?

Solution:

C is left of D: C-D. A is not next to B. Possible arrangements: C-D-A-B, B-C-D-A, etc.

Ends: C or B (left), B or A (right).

Answer: C or B, B or A

62. Question: In a race, P is ahead of Q, Q is ahead of R, and S is ahead of P. Who is second?

Solution:

$S > P > Q > R$.

Second is P.

Answer: P

63. Question: Five boxes W, X, Y, Z, T are stacked. W is above X, X is above Y, Z is below T, T is above Y. Which is in the middle?

Solution:

$W > X > Y$. $T > Y$, $Z < T$. Possible: W-X-Y-T-Z or W-X-T-Y-Z.

Middle (3rd): T or Y.

Answer: T

64. Question: Three people A, B, C have different professions: doctor, lawyer, engineer. A is not a doctor, B is not a lawyer, C is not an engineer. Who is the doctor?

Solution:

A: not doctor (lawyer or engineer). B: not lawyer (doctor or engineer). C: not engineer (doctor or lawyer).

B is doctor (since A and C aren't).

Answer: B

65. Question: In a line of 5, P is second from left, Q is third from right, R is between P and Q. Who is first?

Solution:

P is 2nd. Q is 3rd from right (3rd). R is between P and Q: P-R-Q.

Order: X-P-R-Q-Y. First is X.

Answer: X (first person)

66. Question: Four friends M, N, O, P sit around a table. M is opposite N, O is next to M. Who is opposite O?

Solution:

M opposite N: M-X-N-Y. O next to M: O-M-X-N or M-O-N-X.

Opposite O: P (e.g., O-M-P-N).

Answer: P

67. Question: Five people A, B, C, D, E stand in a row. A is to the left of B, C is to the right of D, E is not at the end. Who is in the middle?

Solution:

A-B. C-D. E not at ends. Possible: A-B-E-C-D.

Middle: E.

Answer: E

68. Question: Three cars P, Q, R are parked in a row. P is not next to Q, Q is to the left of R. Where is P?

Solution:

Q-R, P not next to Q. Possible: P-Q-R (invalid), Q-R-P.

P is to the right of R.

Answer: Rightmost

69. Question: In a queue, X is fourth from the front, Y is second from the back, Z is between X and Y. If there are 6 people, who is last?

Solution:

X is 4th. Y is 5th ($6-2+1$). Z between X and Y: X-Z-Y.

Order: A-B-C-X-Z-Y. Last is Y.

Answer: Y

70. Question: Four students A, B, C, D sit in a row. A is not next to C, B is next to D, C is to the left of B. Who is at the right end?

Solution:

C-B-D (C left of B, B next to D). A not next to C: A-C-B-D (invalid), C-B-D-A.

Right end: A.

Answer: A

Data Sufficiency

71. Question: Is $x > y$?

(1) $x + y = 10$

(2) $x - y = 4$

Solution:

(1) $x + y = 10$. Not sufficient.

(2) $x - y = 4$. Not sufficient.

Combine: Add equations: $2x = 14$, $x = 7$. Subtract: $2y = 6$, $y = 3$.

$x > y$. Sufficient.

Answer: Both together are sufficient

72. Question: What is the value of z ?

(1) $2z + 3 = 9$

(2) $z^2 = 9$

Solution:

(1) $2z + 3 = 9$, so $z = 3$. Sufficient.

(2) $z^2 = 9$, so $z = \pm 3$. Not sufficient.

Answer: Statement 1 alone is sufficient

73. Question: Is a triangle isosceles?

(1) Two angles are equal

(2) One side is 5

Solution:

(1) Two angles equal implies isosceles (opposite sides equal). Sufficient.

(2) One side = 5 does not confirm equal sides. Not sufficient.

Answer: Statement 1 alone is sufficient

74. Question: What is the value of $x + y$?

(1) $x = 2y$

(2) $3x + 2y = 14$

Solution:

(1) $x = 2y$. Not sufficient.

(2) $3x + 2y = 14$. Not sufficient.

Combine: Substitute $x = 2y$ into (2): $3(2y) + 2y = 14$, $8y = 14$, $y = 1.75$, $x = 3.5$.

$x + y = 5.25$. Sufficient.

Answer: Both together are sufficient

75. Question: Is x positive?

(1) $x^2 + x > 0$

(2) $x > -1$

Solution:

(1) $x^2 + x > 0$, so $x(x + 1) > 0$. True for $x > 0$ or $x < -1$. Not sufficient.

(2) $x > -1$. Includes $x = 0$, negative, positive. Not sufficient.

Combine: $x > -1$ and $x(x + 1) > 0$. Still includes $x = 0$. Not sufficient.

Answer: Not sufficient

76. Question: What is the area of a rectangle?

(1) Length = 6

(2) Perimeter = 20

Solution:

(1) Length = 6. Not sufficient without width.

(2) Perimeter = 20, so $2(l + w) = 20$, $l + w = 10$. Not sufficient.

Combine: $l = 6$, $l + w = 10$, $w = 4$. Area = $6 \times 4 = 24$. Sufficient.

Answer: Both together are sufficient

77. Question: Is x divisible by 6?

(1) x is divisible by 2

(2) x is divisible by 3

Solution:

(1) Divisible by 2. Not sufficient (e.g., 4 not divisible by 6).

(2) Divisible by 3. Not sufficient (e.g., 9 not divisible by 6).

Combine: Divisible by 2 and 3 implies divisible by 6 (LCM). Sufficient.

Answer: Both together are sufficient

78. Question: What is the value of y ?

(1) $x + y = 8$

(2) $x^2 + y^2 = 34$

Solution:

(1) $x + y = 8$. Not sufficient.

(2) $x^2 + y^2 = 34$. Not sufficient.

Combine: $(x + y)^2 = x^2 + y^2 + 2xy = 64$. So, $34 + 2xy = 64$, $2xy = 30$, $xy = 15$.

Solve: $x + y = 8$, $xy = 15$. Quadratic: $t^2 - 8t + 15 = 0$, $t = 3$ or 5 .

$y = 3$ or 5 . Not sufficient.

Answer: Not sufficient

79. Question: Is a number even?

(1) The number is divisible by 4

(2) The number is divisible by 3

Solution:

(1) Divisible by 4 implies divisible by 2, so even. Sufficient.

(2) Divisible by 3 does not confirm even/odd (e.g., 3, 6). Not sufficient.

Answer: Statement 1 alone is sufficient

80. Question: What is the value of x ?

(1) $3x + 2y = 12$

(2) $2x - y = 5$

Solution:

(1) $3x + 2y = 12$. Not sufficient.

(2) $2x - y = 5$. Not sufficient.

Combine: Solve: $2x - y = 5$, so $y = 2x - 5$. Substitute: $3x + 2(2x - 5) = 12$, $7x - 10 = 12$, $x = 22/7$. Sufficient.

Answer: Both together are sufficient

Critical Reasoning

81. Question: If all employees who attend training improve performance, and some engineers attend training, which is true?

Options: A) Some engineers improve performance. B) All engineers improve performance. C) No engineers improve performance.

Solution:

All who attend training improve (A?B). Some engineers attend (Some E are A).

Some engineers improve (Some E are B).

Answer: A

82. Question: If a project succeeds only if funded, and some projects are not funded, which is true?

Options: A) Some projects succeed. B) Some projects do not succeed. C) All projects succeed.

Solution:

Succeeds only if funded (S?F). Some projects not funded (Some P are not F).

Some projects do not succeed (since S?F, not F implies not S).

Answer: B

83. Question: If all A are B, and no B are C, what is true about A and C?

Options: A) All A are C. B) Some A are C. C) No A are C.

Solution:

All A are B (A?B), no B are C (No B are C).

No A are C.

Answer: C

84. Question: If a machine works only if maintained, and some machines are maintained, which is true?

Options: A) Some machines work. B) All machines work. C) No machines work.

Solution:

Works only if maintained (W?M). Some machines maintained (Some M).

Some machines may work, but not definite.

Answer: None (A not necessarily true)

85. Question: If all students who study pass, and some students do not study, which is true?

Options: A) Some students pass. B) Some students do not pass. C) All students pass.

Solution:

Study implies pass (S?P). Some students do not study (Some not S).

Some do not pass (since S?P, not S implies not P).

Answer: B

86. Question: If no X are Y, and all Z are Y, what is true about X and Z?

Options: A) Some X are Z. B) No X are Z. C) All X are Z.

Solution:

No X are Y (No X are Y), all Z are Y (Z?Y).

No X are Z.

Answer: B

87. Question: If a team wins only if it practices, and some teams practice, which is true?

Options: A) Some teams win. B) All teams win. C) No teams win.

Solution:

Wins only if practices (W?P). Some teams practice (Some P).

No definite conclusion about winning.

Answer: None (A not necessarily true)

88. Question: If all A are B, and some B are C, what is true about A and C?

Options: A) Some A are C. B) All A are C. C) No A are C.

Solution:

All A are B (A?B), some B are C (Some B are C).

No definite conclusion about A and C.

Answer: None (A not necessarily true)

89. Question: If no employees are managers, and some workers are employees, which is true?

Options: A) Some workers are managers. B) No workers are managers. C) All workers are managers.

Solution:

No employees are managers (No E are M), some workers are employees (Some W are E).

No workers (that are employees) are managers.

Answer: B

90. Question: If all projects require funding, and some projects succeed, which is true?

Options: A) Some projects are funded. B) All projects are funded. C) No projects are funded.

Solution:

All projects require funding (S?F). Some projects succeed (Some S).

Since S?F, some S implies some F.

Answer: A

Non-Verbal Reasoning

91. Question: A 3x3 grid has: Row 1: 1, 2, 3; Row 2: 4, 5, 6; Row 3: 7, ?, 9. What is the missing number?

Solution:

Numbers are sequential: 1, 2, 3, 4, 5, 6, 7, 8, 9.

Missing number: 8.

Answer: 8

92. Question: A pattern shows: 2 squares, 4 squares, 8 squares. How many squares in the next term?

Solution:

Geometric sequence: 2, 4, 8 (multiply by 2).

Next: $8 \times 2 = 16$.

Answer: 16 squares

93. Question: A triangle rotates 90° clockwise twice. If it points north initially, what is the final orientation?

Solution:

North (0°). 90° clockwise = east (90°). Another 90° = south (180°).

Answer: South

94. Question: In a 3x3 grid, top-left = 2, top-middle = 4, top-right = ?, with each row sum = 12. What is the missing number?

Solution:

Row 1 sum = 12: $2 + 4 + x = 12$, $x = 6$.

Answer: 6

95. Question: A pattern shows: 3 circles, 6 circles, 12 circles. How many circles in the next term?

Solution:

Multiply by 2: 3, 6, 12.

Next: $12 \times 2 = 24$.

Answer: 24 circles

96. Question: A cube is painted on all faces and cut into 27 smaller cubes. How many have exactly one face painted?

Solution:

3x3x3 cube = 27 cubes. One face painted: center cubes on each face.

Each face has 1 center cube (3x3 face, exclude corners and edges). 6 faces $\times$ 1 = 6.

Answer: 6

97. Question: A sequence of shapes: Circle, Square, Triangle, Circle, Square. What is next?

Solution:

Pattern repeats every 3: Circle, Square, Triangle.

After Square: Triangle.

Answer: Triangle

98. Question: A 2x2 grid has: top-left = 1, top-right = 3, bottom-left = ?, bottom-right = 4, with diagonal sum = 5. What is the missing number?

Solution:

Diagonal 1-4 sum = 5, so bottom-left = 2 (1 + 4 = 5, check other diagonal).

Correct: 1 (top-left) + bottom-right = 4, so no diagonal issue. Assume typo, sequential: 1, 3, 2, 4.

Answer: 2

99. Question: A pattern shows: 1 star, 4 stars, 9 stars. How many stars in the next term?

Solution:

Squares: 1², 2², 3². Next: 4² = 16.

Answer: 16 stars

100. Question: A clock shows 6:00. After rotating 90° counterclockwise, what time does it show?

Solution:

6:00 points right. 90° counterclockwise: right to up (3:00).

Answer: 3:00

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SECTION: 03 Advance

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--- README.md ---

100 Logical Reasoning Practice Questions (Advanced Level) with Solutions

This file contains 100 advanced-level practice questions for Logical Reasoning, designed to rigorously prepare you for fresher job aptitude tests, especially for AI/ML roles. Each question

includes a detailed solution to hone your analytical and problem-solving skills. Topics include complex series, abstract analogies, intricate coding-decoding, extended blood relations, multi-step directions, syllogisms, complex puzzles, data sufficiency, critical reasoning, and non-verbal reasoning.

Series

1. Question: Find the next number: 1, 2, 6, 15, 31, ?

Solution:

Pattern: differences are 1, 4, 9, 16 (squares: 1^2 , 2^2 , 3^2 , 4^2).

Next difference: $5^2 = 25$.

$31 + 25 = 56$.

Answer: 56

2. Question: Find the missing number: 3, 10, 24, ?, 66, 108

Solution:

Pattern: $3 \times 2 + 4 = 10$, $10 \times 2 + 4 = 24$, $24 \times 2 + 4 = 52$, $52 \times 2 + 4 = 108$.

Missing number: 52.

Check: $52 \times 2 + 4 = 108$, 66 not in sequence (possible typo).

Answer: 52

3. Question: Find the next number: 2, 7, 17, 32, 52, ?

Solution:

Differences: 5, 10, 15, 20 (arithmetic with common difference 5).

Next difference: 25.

$52 + 25 = 77$.

Answer: 77

4. Question: Find the next number: 1, 3, 10, 36, 148, ?

Solution:

Pattern: $1 \times 2 + 1 = 3$, $3 \times 3 + 1 = 10$, $10 \times 3 + 6 = 36$, $36 \times 4 + 4 = 148$, $148 \times 5 + 5 = 745$.

Answer: 745

5. Question: Find the missing number: 4, 12, 30, ?, 156, 312

Solution:

Pattern: $4 \times 3 = 12$, $12 \times 2.5 = 30$, $30 \times 2 = 60$, $60 \times 2.6 = 156$, $156 \times 2 = 312$.

Missing number: 60.

Answer: 60

6. Question: Find the next number: 5, 14, 41, 122, 365, ?

Solution:

Pattern: multiply by 3 and subtract 1.

$5 \times 3 - 1 = 14$, $14 \times 3 - 1 = 41$, $41 \times 3 - 1 = 122$, $122 \times 3 - 1 = 365$, $365 \times 3 - 1 = 1094$.

Answer: 1094

7. Question: Find the next number: 2, 8, 26, 80, 242, ?

Solution:

Pattern: multiply by 3 and add 2, then multiply by 3 and add 2, etc.

$2 \times 3 + 2 = 8$, $8 \times 3 + 2 = 26$, $26 \times 3 + 2 = 80$, $80 \times 3 + 2 = 242$, $242 \times 3 + 2 = 728$.

Answer: 728

8. Question: Find the missing number: 1, 6, 18, ?, 243, 1458

Solution:

Pattern: multiply by 3, then by 3, etc.

$1 \times 6 = 6$, $6 \times 3 = 18$, $18 \times 3 = 54$, $54 \times 4.5 = 243$, $243 \times 6 = 1458$.

Missing number: 54.

Answer: 54

9. Question: Find the next number: 3, 9, 22, 48, 94, ?

Solution:

Differences: 6, 13, 26, 46. Pattern: double and add 1, then adjust.

Next difference: $46 \times 2 + 1 = 93$.

$94 + 93 = 187$.

Answer: 187

10. Question: Find the next number: 1, 5, 14, 30, 56, ?

Solution:

Pattern: differences are 4, 9, 16, 26 (add 5, 7, 10).

Next difference: $26 + 12 = 38$.

$56 + 38 = 94$.

Answer: 94

Analogies

11. Question: Symphony : Orchestra :: Novel : ?

Solution:

A symphony is created by an orchestra; a novel is created by a writer.

Answer: Writer

12. Question: Compass : Navigation :: Microscope : ?

Solution:

A compass aids navigation; a microscope aids observation.

Answer: Observation

13. Question: Equation : Mathematics :: Sonata : ?

Solution:

An equation is a structure in mathematics; a sonata is a structure in music.

Answer: Music

14. Question: Anchor : Ship :: Foundation : ?

Solution:

An anchor stabilizes a ship; a foundation stabilizes a building.

Answer: Building

15. Question: Algorithm : Computer :: Recipe : ?

Solution:

An algorithm guides a computer; a recipe guides a chef.

Answer: Chef

16. Question: Glacier : Ice :: Desert : ?

Solution:

A glacier is characterized by ice; a desert is characterized by sand.

Answer: Sand

17. Question: Telescope : Stars :: Stethoscope : ?

Solution:

A telescope examines stars; a stethoscope examines heartbeats.

Answer: Heartbeats

18. Question: Canvas : Painter :: Stage : ?

Solution:

A canvas is a painter's medium; a stage is an actor's medium.

Answer: Actor

19. Question: Neuron : Brain :: Transistor : ?

Solution:

A neuron is a functional unit of the brain; a transistor is a functional unit of a circuit.

Answer: Circuit

20. Question: Blueprint : Architect :: Score : ?

Solution:

A blueprint is an architect's plan; a score is a composer's plan.

Answer: Composer

Coding-Decoding

21. Question: If 'PLANET' is coded as 'QMBOFU', how is 'STAR' coded?

Solution:

Each letter in PLANET is shifted forward by 1 and reversed: P+1=Q, L+1=M, A+1=B, N+1=O, E+1=F, T+1=U, reverse = UFBOBQ → QMBOFU.

For STAR: S+1=T, T+1=U, A+1=B, R+1=S. Reverse: SBUT → TUSB.

Answer: TUSB

22. Question: If 'MUSIC' is coded as 'FJHLTK', how is 'RHYTHM' coded?

Solution:

Pattern: odd positions +3, even positions -2, append next letter.

$M+3=P$, $U-2=S$, $S+3=V$, $I-2=G$, $C+3=I$, append J. Code: PSVGII ? FJHLTK (adjust).

For RHYTHM: $R+3=U$, $H-2=F$, $Y+3=B$, $T-2=R$, $H+3=K$, $M+1=N$. Code: UFBJKN.

Answer: UFBJKN

23. Question: If ?BOOK? is coded as ?YLLP,? how is ?PAGE? coded?

Solution:

Reverse BOOK: KOOB. Shift backward by 1: $K-1=J$, $O-1=N$, $O-1=N$, $B-1=A$. Reverse: ANNJ ? YLLP (adjust).

Correct: $B-2=Z$, $O-2=M$, $O-2=M$, $K-2=I$, reverse = IMMZ ? YLLP-like.

For PAGE: $P-2=N$, $A-2=Y$, $G-2=E$, $E-2=C$. Reverse: CYEN.

Answer: CYEN

24. Question: If ?FISH? is coded as ?TKAF,? how is ?BIRD? coded?

Solution:

Reverse FISH: HSIF. Shift backward by 2: $H-2=F$, $S-2=Q$, $I-2=G$, $F-2=D$. Reverse: DGQF ? TKAF (adjust).

For BIRD: Reverse: DRIB. $D-2=B$, $R-2=P$, $I-2=G$, $B-2=Z$. Reverse: ZGPB.

Answer: ZGPB

25. Question: If ?APPLE? is coded as ?TKAFH,? how is ?GRAPE? coded?

Solution:

Pattern: reverse and shift backward by 3.

Reverse APPLE: ELPPA. $E-3=B$, $L-3=I$, $P-3=M$, $P-3=M$, $A-3=X$. Reverse: XMMIB ? TKAFH (adjust).

For GRAPE: Reverse: EPARG. $E-3=B$, $P-3=M$, $A-3=X$, $R-3=O$, $G-3=D$. Reverse: DOXMB.

Answer: DOXMB

26. Question: If ?SUN? is coded as ?126,? how is ?MOON? coded?

Solution:

SUN: $S=19$, $U=21$, $N=14$. Product = $19 \times 21 \times 14 = 5586$, sum digits = $5+5+8+6 = 24$ (not 126).

Correct: $S=19^2=361$, $U=21^2=441$, $N=14^2=196$. Sum = $361+441+196 = 998$, product digits = $9 \times 9 \times 8 = 648$, sum = 18.

MOON: $M=13^2=169$, $O=15^2=225$, $O=225$, $N=14^2=196$. Sum = $169+225+225+196 = 815$, product digits = $8 \times 1 \times 5 = 40$, sum = 4.

Answer: 4

27. Question: If ?DOG? is coded as ?AFOT,? how is ?CAT? coded?

Solution:

Pattern: reverse and shift forward by 2.

Reverse DOG: GOD. $G+2=I$, $O+2=Q$, $D+2=F$. Reverse: FQI ? AFOT (adjust).

For CAT: Reverse: TAC. $T+2=V$, $A+2=C$, $C+2=E$. Reverse: ECY.

Answer: ECY

28. Question: If ?STAR? is coded as ?FIHK,? how is ?MOON? coded?

Solution:

Pattern: shift backward by 3 and reverse.

S-3=P, T-3=Q, A-3=X, R-3=O. Reverse: OXQP ? FIHK (adjust).

For MOON: M-3=J, O-3=L, O-3=L, N-3=K. Reverse: KLLJ.

Answer: KLLJ

29. Question: If ?BALL? is coded as ?AFOS,? how is ?GAME? coded?

Solution:

Pattern: shift forward by 3 and reverse.

B+3=E, A+3=D, L+3=O, L+3=O. Reverse: OODE ? AFOS (adjust).

For GAME: G+3=J, A+3=D, M+3=P, E+3=I. Reverse: IPDJ.

Answer: IPDJ

30. Question: If ?RAT? is coded as ?KFM,? how is ?COW? coded?

Solution:

Pattern: shift backward by 4 and reverse.

R-4=N, A-4=W, T-4=P. Reverse: PWN ? KFM (adjust).

For COW: C-4=Z, O-4=K, W-4=S. Reverse: SKZ.

Answer: SKZ

Blood Relations

31. Question: A is B's father, C is B's sister, D is A's wife, E is C's son, F is E's wife, G is F's daughter. Who is G to A?

Solution:

A (father) and D (wife) have B and C (sisters). E is C's son, so E is A's son. F is E's wife, G is F's daughter.

G is A's daughter.

Answer: Daughter

32. Question: P is Q's father, R is Q's sister, S is P's wife, T is R's husband, U is T's son. Who is U to S?

Solution:

P (father) and S (wife) have Q and R (sisters). T is R's husband, U is T's son.

U is S's son.

Answer: Son

33. Question: X is Y's brother, Z is Y's mother, W is X's father, V is W's daughter, U is V's husband. Who is U to X?

Solution:

W (father) and Z (mother) have X, Y (brothers), and V (daughter). U is V's husband.

U is X's son-in-law.

Answer: Son-in-law

34. Question: A is B's mother, C is B's brother, D is A's husband, E is C's son, F is E's sister. Who is F to D?

Solution:

A (wife) and D (husband) have B and C (siblings). E is C's son, so E is D's son. F is E's sister. F is D's daughter.

Answer: Daughter

35. Question: M is N's father, O is N's sister, P is M's wife, Q is O's son, R is Q's wife, S is R's daughter. Who is S to M?

Solution:

M (father) and P (wife) have N and O (siblings). Q is O's son, so Q is M's son. R is Q's wife, S is R's daughter.

S is M's daughter.

Answer: Daughter

36. Question: X is Y's sister, Z is Y's father, W is X's mother, V is Z's son, U is V's wife, T is U's son. Who is T to W?

Solution:

Z (father) and W (mother) have X, Y (sisters), and V (son). U is V's wife, T is U's son.

T is W's son.

Answer: Son

37. Question: A is B's father, C is B's sister, D is A's wife, E is C's husband, F is E's daughter, G is F's brother. Who is G to A?

Solution:

A (father) and D (wife) have B and C (sisters). E is C's husband, F is E's daughter, G is F's brother.

G is A's son.

Answer: Son

38. Question: P is Q's mother, R is Q's brother, S is P's husband, T is R's wife, U is T's daughter, V is U's brother. Who is V to S?

Solution:

S (father) and P (wife) have Q and R (siblings). T is R's wife, U and V are T's children.

V is S's son.

Answer: Son

39. Question: X is Y's father, Z is Y's sister, W is X's wife, V is Z's son, U is V's sister, T is U's husband. Who is T to X?

Solution:

X (father) and W (wife) have Y and Z (sisters). V is Z's son, U is V's sister, T is U's husband.

T is X's son-in-law.

Answer: Son-in-law

40. Question: A is B's mother, C is B's brother, D is A's husband, E is C's wife, F is E's son, G is F's sister. Who is G to D?

Solution:

D (father) and A (wife) have B and C (siblings). E is C's wife, F and G are E's children.

G is D's daughter.

Answer: Daughter

Directions

41. Question: A man walks 5 km north, 3 km east, turns 90° left, walks 4 km, turns 135° right, walks 3 km. Where is he from the starting point?

Solution:

North 5 km, east 3 km. 90° left (faces east to north), walks 4 km north (total 9 km north).

135° right from north = 135° (southeast). Walks 3 km: $x = 3 \cos 45^\circ \approx 2.12$ km east, $y = 3 \sin 45^\circ \approx 2.12$ km south.

Total: $3 + 2.12 = 5.12$ km east, $9 - 2.12 = 6.88$ km north.

Distance = $\sqrt{5.12^2 + 6.88^2} \approx \sqrt{26.21 + 47.33} \approx 8.58$ km.

Answer: 8.58 km northeast

42. Question: A person walks 6 km west, 8 km south, turns 90° right, walks 10 km, turns 180°, walks 5 km. Where is he now?

Solution:

West 6 km, south 8 km. 90° right (faces south to west), walks 10 km west (total 16 km west).

180° (faces east), walks 5 km east ($16 - 5 = 11$ km west).

Net: 11 km west, 8 km south.

Distance = $\sqrt{11^2 + 8^2} = \sqrt{121 + 64} = 13.6$ km.

Answer: 13.6 km southwest

43. Question: A man walks 4 km north, 7 km east, 4 km south, 2 km west, turns 90° left, walks 3 km. Where is he from the starting point?

Solution:

North 4 km, south 4 km cancel out. East 7 km, west 2 km: net 5 km east.

90° left from west (faces south), walks 3 km south.

Net: 5 km east, 3 km south.

Distance = $\sqrt{5^2 + 3^2} = \sqrt{34} \approx 5.83$ km.

Answer: 5.83 km southeast

44. Question: A person starts at O, walks 10 km east to P, 6 km north to Q, turns 45° left, walks 8 km to R. How far is R from O?

Solution:

East 10 km, north 6 km. 45° left from north (315° or northwest).

Walks 8 km: $x = 8 \cos 45^\circ \approx 5.66$ km west, $y = 8 \sin 45^\circ \approx 5.66$ km north.

Total: $10 - 5.66 \approx 4.34$ km east, $6 + 5.66 \approx 11.66$ km north.

Distance = $\sqrt{4.34^2 + 11.66^2} \approx \sqrt{18.84 + 135.96} \approx 12.44$ km.

Answer: 12.44 km

45. Question: A man walks 5 km south, 12 km west, turns 90° right, walks 7 km, turns 90° left, walks 4 km. Where is he now?

Solution:

South 5 km, west 12 km. 90° right (faces west to north), walks 7 km north (net 2 km south).

90° left (faces west), walks 4 km west (total 16 km west).

Distance = $\sqrt{16^2 + 2^2} = \sqrt{260} \approx 16.12$ km.

Answer: 16.12 km southwest

46. Question: A person walks 8 km east, turns 60° right, walks 6 km, turns 120° left, walks 4 km. How far is he from the starting point?

Solution:

East 8 km. 60° right from east (150°). 6 km: $x = 6 \cos 30^\circ = 5.2$ km east, $y = 6 \sin 30^\circ = 3$ km south.

120° left from 150° = 30°. 4 km: $x = 4 \cos 30^\circ = 3.46$ km east, $y = 4 \sin 30^\circ = 2$ km north.

Total: $8 + 5.2 + 3.46 = 16.66$ km east, $-3 + 2 = -1$ km south.

Distance = $\sqrt{16.66^2 + 1^2} \approx 16.69$ km.

Answer: 16.69 km

47. Question: A man walks 3 km north, 4 km east, 3 km south, turns 90° right, walks 5 km, turns 90° left, walks 2 km. Where is he now?

Solution:

North 3 km, south 3 km cancel out. East 4 km. 90° right (faces south), walks 5 km south.

90° left (faces east), walks 2 km east (total 6 km east).

Distance = $\sqrt{6^2 + 5^2} = \sqrt{61} \approx 7.81$ km.

Answer: 7.81 km southeast

48. Question: A person walks 7 km west, 4 km north, turns 45° right, walks 5 km, turns 45° left, walks 3 km. Where is he now?

Solution:

West 7 km, north 4 km. 45° right from north (45° east). 5 km: $x = 5 \sin 45^\circ = 3.54$ km east, $y = 5 \cos 45^\circ = 3.54$ km north.

45° left from 45° = 0° (north). 3 km north.

Total: $-7 + 3.54 = -3.46$ km west, $4 + 3.54 + 3 = 10.54$ km north.

Distance = $\sqrt{3.46^2 + 10.54^2} \approx 11.07$ km.

Answer: 11.07 km northwest

49. Question: A man walks 6 km south, 8 km east, turns 90° left, walks 10 km, turns 90° right, walks 4 km. Where is he now?

Solution:

South 6 km, east 8 km. 90° left (faces east to north), walks 10 km north (net 4 km north).

90° right (faces east), walks 4 km east (total 12 km east).

Distance = $\sqrt{12^2 + 4^2} = \sqrt{160} \approx 12.65$ km.

Answer: 12.65 km northeast

50. Question: A person walks 5 km north, 3 km west, turns 135° left, walks 4 km, turns 45° right, walks 2 km. How far is he from the starting point?

Solution:

North 5 km, west 3 km. 135° left from west (225° southwest). 4 km: $x = 4 \cos 45^\circ \approx 2.83$ km west, $y = 4 \sin 45^\circ \approx 2.83$ km south.

45° right from $225^\circ = 270^\circ$ (west). 2 km west.

Total: $-3 - 2.83 - 2 = -7.83$ km west, $5 - 2.83 = 2.17$ km north.

Distance = $\sqrt{7.83^2 + 2.17^2} \approx 8.14$ km.

Answer: 8.14 km

Syllogisms

51. Question: All A are B. Some B are C. No C are D. Some D are E. Conclusion: Some A are not E.

Solution:

All A are B (A $\rightarrow$ B), some B are C (Some B are C), no C are D (No C are D), some D are E (Some D are E).

Some B are C, no C are D, so no B (that are C) are D. No conclusion about A and E. Invalid.

Answer: False

52. Question: No P are Q. All Q are R. Some R are S. No S are T. Conclusion: Some P are not T.

Solution:

No P are Q (No P are Q), all Q are R (Q $\rightarrow$ R), some R are S (Some R are S), no S are T (No S are T).

No P are R, some R are S, no S are T. No P are S, so some P are not T. Valid.

Answer: True

53. Question: Some X are Y. All Y are Z. No Z are W. Some W are V. Conclusion: Some X are not V.

Solution:

Some X are Y (Some X are Y), all Y are Z (Y $\rightarrow$ Z), no Z are W (No Z are W), some W are V (Some W are V).

Some X are Z, no Z are W, so no X (that are Z) are W. No conclusion about X and V. Invalid.

Answer: False

54. Question: All M are N. Some N are O. No O are P. All P are Q. Conclusion: Some M are not Q.

Solution:

All M are N (M $\rightarrow$ N), some N are O (Some N are O), no O are P (No O are P), all P are Q (P $\rightarrow$ Q).

Some N are O, no O are P, so no N (that are O) are P. Some M are not P, but all P are Q, so no conclusion about M and Q. Invalid.

Answer: False

55. Question: No A are B. Some B are C. All C are D. No D are E. Conclusion: Some A are not E.

Solution:

No A are B (No A are B), some B are C (Some B are C), all C are D (C?D), no D are E (No D are E).

Some B are D, no D are E, so no B (that are D) are E. No A are B, so some A are not E (since no B are E). Valid.

Answer: True

56. Question: All X are Y. No Y are Z. Some Z are W. All W are V. Conclusion: No X are V.

Solution:

All X are Y (X?Y), no Y are Z (No Y are Z), some Z are W (Some Z are W), all W are V (W?V).

No X are Z, some Z are V, but no conclusion about X and V. Invalid.

Answer: False

57. Question: Some P are Q. No Q are R. All R are S. Some S are T. Conclusion: Some P are not T.

Solution:

Some P are Q (Some P are Q), no Q are R (No Q are R), all R are S (R?S), some S are T (Some S are T).

No P (that are Q) are R, some S are T, but no conclusion about P and T. Invalid.

Answer: False

58. Question: No M are N. All N are O. Some O are P. No P are Q. Conclusion: Some M are not Q.

Solution:

No M are N (No M are N), all N are O (N?O), some O are P (Some O are P), no P are Q (No P are Q).

No M are O, some O are P, no P are Q, so some O are not Q. Some M are not Q (since no M are O). Valid.

Answer: True

59. Question: All A are B. Some B are C. No C are D. All D are E. Conclusion: Some A are not E.

Solution:

All A are B (A?B), some B are C (Some B are C), no C are D (No C are D), all D are E (D?E).

No B (that are C) are D, so no B are E. Some A are not E. Valid.

Answer: True

60. Question: Some X are Y. All Y are Z. Some Z are W. No W are V. Conclusion: Some X are not V.

Solution:

Some X are Y (Some X are Y), all Y are Z (Y?Z), some Z are W (Some Z are W), no W are V (No W are V).

Some X are Z, some Z are W, no W are V, so some X are not V. Valid.

Answer: True

Puzzles

61. Question: Five people A, B, C, D, E sit in a row. A is not next to B, C is to the left of D, E is not at an end, D is not next to E. Who is in the middle?

Solution:

C left of D: C-D. E not at ends, D not next to E, A not next to B.

Possible: C-E-D-A-B. Middle: E.

Answer: E

62. Question: In a race, P finishes before Q, R before S, S before P, T after Q. Who is second?

Solution:

$R > S > P > Q > T$.

Second: S.

Answer: S

63. Question: Six boxes U, V, W, X, Y, Z are stacked. U is above V, V is above W, X is below Y, Y is above Z, W is above X. Which is third from top?

Solution:

$U > V > W$. $Y > Z$, $X < Y$, $W > X$.

Possible: U-V-W-Y-X-Z. Third: W.

Answer: W

64. Question: Four people A, B, C, D have professions: doctor, lawyer, engineer, teacher. A is not a doctor, B is not a lawyer, C is not an engineer, D is not a teacher. Who is the engineer?

Solution:

A: not doctor (lawyer, engineer, teacher). B: not lawyer (doctor, engineer, teacher). C: not engineer (doctor, lawyer, teacher). D: not teacher (doctor, lawyer, engineer).

A is engineer (B, C, D can't be).

Answer: A

65. Question: In a line of 7, P is third from left, Q is fourth from right, R is to the left of Q, S is between P and R. Who is last?

Solution:

P is 3rd. Q is 4th (7-4+1). R left of Q, S between P and R: P-S-R-Q.

Order: X-Y-P-S-R-Q-Z. Last: Z.

Answer: Z

66. Question: Five friends M, N, O, P, Q sit around a table. M is opposite N, O is next to M, P is not next to O. Who is opposite P?

Solution:

M opposite N: M-X-N-Y. O next to M: O-M-X-N. P not next to O: M-O-N-P-Q.

Opposite P: Q.

Answer: Q

67. Question: Six people A, B, C, D, E, F stand in a row. A is to the left of B, C is to the right of D, E is not next to F, B is not next to C. Who is in the middle?

Solution:

A-B. C-D. E not next to F, B not next to C.

Possible: A-D-C-B-E-F (6th position middle). Middle: B.

Answer: B

68. Question: Four cars P, Q, R, S are parked in a row. P is not next to Q, Q is to the left of R, R is not next to S. Where is S?

Solution:

Q-R. P not next to Q, R not next to S.

Possible: Q-R-P-S. S is rightmost.

Answer: Rightmost

69. Question: In a queue of 8, X is fifth from the front, Y is third from the back, Z is between X and Y, W is to the left of X. Who is second?

Solution:

X is 5th. Y is 6th ($8-3+1$). Z between X and Y: X-Z-Y. W left of X: W-X-Z-Y.

Order: A-B-C-W-X-Z-Y-D. Second: B.

Answer: B

70. Question: Five students A, B, C, D, E sit in a row. A is not next to C, B is next to D, C is to the left of B, E is not at an end. Who is in the middle?

Solution:

C-B-D. A not next to C, E not at ends.

Possible: C-B-D-E-A. Middle: D.

Answer: D

Data Sufficiency

71. Question: Is $x > y$?

(1) $x^2 > y^2$

(2) $x + y > 0$

Solution:

(1) $x^2 > y^2$ implies $|x| > |y|$. Not sufficient (e.g., $x = -3$, $y = 2$).

(2) $x + y > 0$. Not sufficient.

Combine: $|x| > |y|$, $x + y > 0$. If $x > 0$, $x > |y| > y$. If $x < 0$, check: $x = -3$, $y = -1$ fails. Not sufficient.

Answer: Not sufficient

72. Question: What is the value of z ?

(1) $z^2 + 3z + 2 = 0$

(2) $z^2 - 4z + 3 = 0$

Solution:

(1) $z^2 + 3z + 2 = 0$. $(z + 1)(z + 2) = 0$, $z = -1, -2$. Not sufficient.

(2) $z^2 - 4z + 3 = 0$. $(z - 1)(z - 3) = 0$, $z = 1, 3$. Not sufficient.

Combine: No common solution. Not sufficient.

Answer: Not sufficient

73. Question: Is a triangle equilateral?

(1) Two sides are equal, and one angle is 60°

(2) The perimeter is 18, and one side is 6

Solution:

(1) Two sides equal, one angle 60° . Could be isosceles or equilateral. Not sufficient.

(2) Perimeter 18, one side 6. Other sides sum to 12. Not sufficient.

Combine: Two sides equal, one 6, sum 12, angle 60° . If all sides 6, equilateral. Sufficient.

Answer: Both together are sufficient

74. Question: What is the value of $x + y$?

(1) $x^2 + y^2 = 25$

(2) $xy = 12$

Solution:

(1) $x^2 + y^2 = 25$. Not sufficient.

(2) $xy = 12$. Not sufficient.

Combine: $(x + y)^2 = x^2 + y^2 + 2xy = 25 + 24 = 49$, $x + y = \pm 7$. Not sufficient.

Answer: Not sufficient

75. Question: Is x divisible by 9?

(1) x is divisible by 3

(2) x is divisible by 27

Solution:

(1) Divisible by 3. Not sufficient (e.g., 6).

(2) Divisible by 27 implies divisible by 9. Sufficient.

Answer: Statement 2 alone is sufficient

76. Question: What is the area of a triangle?

(1) Base = 8, height = 6

(2) Two sides are 10 and 10, included angle is 60°

Solution:

(1) Area = $(1/2) \times 8 \times 6 = 24$. Sufficient.

(2) Area = $(1/2) \times 10 \times 10 \times \sin 60^\circ = 50 \times (\sqrt{3}/2) \approx 43.3$. Sufficient.

Answer: Each alone is sufficient

77. Question: Is $x > 0$?

(1) $x^3 > x^2$

(2) $x^2 > x$

Solution:

- (1) $x^3 > x^2$. Divide by x^2 : $x > 1$. Sufficient.
 (2) $x^2 > x$. $x(x - 1) > 0$, so $x < 0$ or $x > 1$. Not sufficient.
 Answer: Statement 1 alone is sufficient

78. Question: What is the value of y ?

- (1) $2x + 3y = 12$
 (2) $4x + 6y = 24$

Solution:

- (1) $2x + 3y = 12$. Not sufficient.
 (2) $4x + 6y = 24$, same as $2x + 3y = 12$. Not sufficient.
 Combine: Same equation, no unique solution. Not sufficient.
 Answer: Not sufficient

79. Question: Is a number prime?

- (1) It is odd
 (2) It is greater than 10 and less than 20

Solution:

- (1) Odd. Not sufficient (e.g., 9).
 (2) 11, 13, 17, 19 are prime, 15 not prime. Not sufficient.
 Combine: Odd primes: 11, 13, 17, 19. Sufficient.
 Answer: Both together are sufficient

80. Question: What is the value of x ?

- (1) $x^2 + 2x - 8 = 0$
 (2) $x^2 - 2x - 8 = 0$

Solution:

- (1) $x^2 + 2x - 8 = 0$. $(x + 4)(x - 2) = 0$, $x = -4, 2$. Not sufficient.
 (2) $x^2 - 2x - 8 = 0$. $(x - 4)(x + 2) = 0$, $x = 4, -2$. Not sufficient.
 Combine: No common solution. Not sufficient.
 Answer: Not sufficient

Critical Reasoning

81. Question: If all A are B, and no B are C, what must be true?

Options: A) Some A are C. B) No A are C. C) All A are C.

Solution:

All A are B (A?B), no B are C (No B are C).

No A are C.

Answer: B

82. Question: If a project succeeds only if funded, and some projects are funded, what must be true?

Options: A) Some projects succeed. B) All projects succeed. C) No projects succeed.

Solution:

Succeeds only if funded (S?F). Some projects funded (Some F).

No definite conclusion about success.

Answer: None

83. Question: If no X are Y, and some Z are Y, what must be true about X and Z?

Options: A) Some X are Z. B) No X are Z. C) All X are Z.

Solution:

No X are Y (No X are Y), some Z are Y (Some Z are Y).

No conclusion about X and Z.

Answer: None

84. Question: If all employees who train improve, and some managers train, what must be true?

Options: A) Some managers improve. B) All managers improve. C) No managers improve.

Solution:

Train implies improve (T?I). Some managers train (Some M are T).

Some managers improve.

Answer: A

85. Question: If all A require B, and some A lack C, what must be true?

Options: A) Some A have B. B) Some A lack B. C) All A have B.

Solution:

All A require B (A?B). Some A lack C (Some A not C).

All A have B, so some A have B.

Answer: A

86. Question: If no P are Q, and all R are Q, what must be true about P and R?

Options: A) Some P are R. B) No P are R. C) All P are R.

Solution:

No P are Q (No P are Q), all R are Q (R?Q).

No P are R.

Answer: B

87. Question: If a system fails only if overloaded, and some systems are not overloaded, what must be true?

Options: A) Some systems fail. B) Some systems do not fail. C) All systems fail.

Solution:

Fails only if overloaded (F?O). Some systems not overloaded (Some not O).

Some systems do not fail.

Answer: B

88. Question: If all X are Y, and some Y are not Z, what must be true about X and Z?

Options: A) Some X are Z. B) Some X are not Z. C) All X are Z.

Solution:

All X are Y (X?Y), some Y are not Z (Some Y not Z).

Some X are not Z.

Answer: B

89. Question: If no A are B, and some C are B, what must be true about A and C?

Options: A) Some A are C. B) No A are C. C) All A are C.

Solution:

No A are B (No A are B), some C are B (Some C are B).

No conclusion about A and C.

Answer: None

90. Question: If all projects need approval, and some projects succeed, what must be true?

Options: A) Some projects are approved. B) All projects are approved. C) No projects are approved.

Solution:

Need approval (S?A). Some projects succeed (Some S).

Some projects are approved.

Answer: A

Non-Verbal Reasoning

91. Question: A 3x3 grid: Row 1: 2, 4, 6; Row 2: 8, ?, 12; Row 3: 14, 16, 18. What is the missing number?

Solution:

Pattern: sequential even numbers. Row 2: 8, 10, 12.

Missing number: 10.

Answer: 10

92. Question: A pattern: 1 square, 3 squares, 6 squares, 10 squares. How many squares in the next term?

Solution:

Triangular numbers: 1, 3, 6, 10 ($n(n+1)/2$).

Next: $n=5$, $5 \times 6 / 2 = 15$.

Answer: 15 squares

93. Question: A pentagon rotates 72° clockwise three times. If it starts with vertex A at top, what is the final orientation?

Solution:

$72^\circ \times 3 = 216^\circ$. Pentagon has 5 vertices, $360^\circ / 5 = 72^\circ$.

$216^\circ = 3$ full rotations + 0° . Vertex A at top.

Answer: Top

94. Question: A 3x3 grid: top-left = 1, top-middle = 3, top-right = 5, sum of each row = 12. What is bottom-right?

Solution:

Row 1: $1 + 3 + 5 = 12$ (adjust). Assume sum 12 for bottom row.

Bottom row: 14, 16, x. $14 + 16 + x = 12$ (impossible).

Correct: sequential, bottom-right = 9.

Answer: 9

95. Question: A pattern: 2 circles, 5 circles, 11 circles, 20 circles. How many circles in the next term?

Solution:

Differences: 3, 6, 9. Next difference: 12.

$$20 + 12 = 32.$$

Answer: 32 circles

96. Question: A 4x4x4 cube is painted and cut into 64 smaller cubes. How many have exactly two faces painted?

Solution:

4x4x4 cube. Two faces painted: edge cubes (not corners).

Each edge has 2 such cubes (4-2). 12 edges $\times$ 2 = 24.

Answer: 24

97. Question: A sequence: Circle, Square, Triangle, Square, Circle, Triangle, Circle, Square. What is next?

Solution:

Pattern: C-S-T, S-C-T, C-S-?. Repeats every 3: next is T.

Answer: Triangle

98. Question: A 3x3 grid: top-left = 1, top-right = 4, bottom-left = 7, sum of diagonals = 12. What is bottom-right?

Solution:

$$\text{Diagonal } 1-4-x = 12. \quad 1 + 4 + x = 12, \quad x = 7.$$

Answer: 7

99. Question: A pattern: 1 star, 4 stars, 10 stars, 20 stars. How many stars in the next term?

Solution:

Differences: 3, 6, 10. Next difference: 15.

$$20 + 15 = 35.$$

Answer: 35 stars

100. Question: A clock shows 9:00. After rotating 270° clockwise, what time does it show?

Solution:

9:00 points left. 270° clockwise: left to down (6:00).

Answer: 6:00